

AMIS 1012

Ethics in Computing



Chapter 3: Organisation Privacy (Part II)



Learning Objectives (Part II)

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1. Identify issues that affect organisation privacy.
2. Propose appropriate strategies to manage and mitigate the impact of computer technology on privacy issues.



Chapter Outline (Part II)

4. Issues Related to Organisation Privacy

4.1 Identity Theft

4.1.1 Data Breaches

4.1.2 Phishing

4.1.3 Deepfake

4.1.4 Spyware

4.2 Workplace Monitoring

4.3 Consumer Profiling

5. Mitigation Measures to address Issues related to Organisation Privacy (eg Cyber Security Preparedness, Measures to Protect Computer Security, Internet Filtering, Firewall)





4. Issues Related to Organisation Privacy

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4.1 Identity Theft

- **Stealing of another person's identity or financial information** in order to commit fraud using that person's identity, such as making false transactions or purchases.
- Outcomes of Identity Theft are **damage to a victim's credit, money, and reputation.**
- **Personal identifying information** include your complete name, home address, email address, Internet username and passwords, Social Security Number (SSN), driver's license number, passport number, or bank number. □ used to conduct **identity theft** or **sell it on the dark Web.**



4. Issues Related to Organisation Privacy

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4.1 Identity Theft (Cont.)

- Few examples of what identity criminals might do with a person's personal identifying information:
 - Use your identification details to **open new credit cards or loans**.
 - Use your name and financial standing to **pass a background check** for a job or to rent an apartment.
 - Use your information to **evade criminal liability**.



4. Issues Related to Organisation Privacy

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4.1 Identity Theft (Cont.)

- How cybercriminal can possibly obtained personal identifying information?
 - Data Breaches
 - Phishing
 - Deepfake
 - Spyware

Note: The above is a non-exhaustive list.



4. Issues Related to Organisation Privacy

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4.1.1 Data Breaches

- A data breach occurs when information is **compromised or removed** from a device **without** the owner's **consent or permission**.
- A data breach may happen to a small business or a big corporation.
- Credit card numbers, consumer records, trade secrets, and national security information are examples of private, proprietary, or classified information that may be stolen.



4. Issues Related to Organisation Privacy

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4.1.1 Data Breaches (Cont.)

- The consequences of a data breach may include **harm** to the target **company's reputation** and face **financial losses**.
- According to the number of data leak events reported between January 2005 and April 2015, the **most leaked record category** was **personally identifiable information**, followed by **financial data**.
- Your classified details could be **traded on the dark Web** after a data breach.

<https://informationisbeautiful.net/visualizations/worlds-biggest-data-breaches-hacks/>



4. Issues Related to Organisation Privacy

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4.1.2 Phishing (Refer to Chapter 2, Phishing)

- Phishing is a technique for **obtaining personal information** through the use of misleading emails and websites.
- The goal is to convince the recipient of the **email** that the letter is something they want or need, such as a bank request or a message from a colleague—and that they may **click on a link or download an attachment**.
- Sender of the emails **pretend to be a trustworthy** human of some kind, typically a genuine or convincing real person or an organization



4. Issues Related to Organisation Privacy

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4.1.3 Deepfake

- Deepfake is an **image or voice processing technique** used to **generate false images and videos**, made possible by advances in artificial intelligence and machine learning technology.
- Email messages aimed at inducing the recipient to pass on confidential information are **much more effective** if they are created by an artificial intelligence system.
- With deepfake, hackers can use **AI-generated videos and voice who seems to be from a genuine person** to obtain personal identifying information via scams using video calls or phone calls.



4. Issues Related to Organisation Privacy

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4.1.4 Spyware

- Spyware is a type of malware that **collects information** about an individual or entity **invisibly** and then distributes it to third parties.
- Advertisers or marketing data companies may be among them, which is why spyware is often referred to as “ **adware**.”
- It is **installed without the user’s knowledge** by strategies such as drive-by downloads, Trojans bundled with legal software, or misleading pop-up windows.

4. Issues Related to Organisation Privacy

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4.1.4 Spyware (Cont.)

- Spyware collects personal information such as your name, address, surfing habits, desires, hobbies, and downloads from your Internet connection.
- A “**keylogger**” is a spyware that captures your email, password, or other credentials.
- Spyware usually **consumes computing resources** such as bandwidth, processing power, and memory.



4. Issues Related to Organisation Privacy

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4.2 Workplace Monitoring

- One of the reasons to implement workplace monitoring is because quite a number of employees **spend significant amounts of time at work performing nonwork-related activities** according to recent studies. Therefore, there are some companies who **banned** the use of **social networking platforms** such as Facebook and Twitter.
- Employers utilize “ **employee tracking**” methods to **monitor their workplaces**, as well as the **locations** and **activities** of their workers.
- These methods include employee monitoring devices, time clocks, video surveillance, GPS services, and biometric technology.



4. Issues Related to Organisation Privacy

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4.2 Workplace Monitoring (Cont.)

- Workplace monitoring can check whether **corporate IT use practices** are being implemented, due to risk of reduced productivity and increased legal liabilities.
- Many businesses in the United States feel it important to **log and monitor employee messages and behaviors** on the job, such as **phone calls, emails, and Web browsing**.
- **Internal fraud** is being avoided, **employee morale** is being assessed, **organisation services** are being appropriately used, and **evidence for any future prosecution** is being gathered.



4. Issues Related to Organisation Privacy

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4.2 Workplace Monitoring (Cont.)

- Prediction and flagging systems that attempt to **forecast employee characteristics or actions**, or that are intended to **detect or prevent** alleged rule-breaking or fraud.
- **Employee biometric and health data** gathered through wearables, fitness monitoring software, and biometric timekeeping systems as part of employer-provided health insurance, workplace wellbeing, and automated job change-tracking tools. Tracking **nonwork-related behaviors and records**, such as health information, **can violate worker privacy**.

4. Issues Related to Organisation Privacy

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4.2 Workplace Monitoring (Cont.)

- Employees, in this case, find that they have **lack of privacy rights**.
- If employees file privacy lawsuits, courts often **rule against workers** who say they were watched when using company devices.
- A private corporation can **defeat a privacy lawsuit** by demonstrating that an **employee was given clear warning** that his or her email, Internet use, and data on company servers were not private and could be tracked.

4. Issues Related to Organisation Privacy

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4.3 Consumer Profiling

- The way to gather the **insights** needed to recognize, segment, and characterize target demographic is through consumer profiling, also known as customer profiling.
- This process begins with an emphasis on existing clients, and moves on to the desired potential demographic and target market to ensure it is the right people for company to reach out to for future promotional/marketing activities.

4. Issues Related to Organisation Privacy

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4.3 Consumer Profiling

- **Methods** to gather consumer's personal information:
 - registration on websites
 - complete surveys
 - fill out paperwork
 - join tournaments online

4. Issues Related to Organisation Privacy

– 4.3 Consumer Profiling

- **Methods** to gather consumer's personal information (Cont.):
 - Collect information on Web users through the use of cookies
 - Cookies are text files that are copied to the hard drives of users who access a website in order for the website to recognize them on future visits.
 - Cookies allow a website to customize the advertisements and promotions it displays to web users.
 - Some cookies will trace the other websites a user has accessed, enabling marketers to make informed assumptions about the types of advertising that will be more appealing to the user
 - use monitoring tools to monitor surfing patterns and deduce personal tastes and desires on their websites.
- Since marketers can gather information about users without their prior consent, the use of cookies and monitoring devices is **debatable**.



5. Mitigation Measures to address Issues related to Organisation Privacy

- **Cyber security preparedness** is where organisations need a cyber defense policy to defend themselves, their clients, and their data from ever-increasing cyber security attacks.



5. Mitigation Measures to address Issues related to Organisation Privacy

5.1 Educate staff about the importance of defense

- Establish common **security procedures and standards** for employees, such as safe passwords and Internet use guidelines that lay out the ramifications of violating the company's cyber security policies.
- Create **policies** for managing and securing customer records and other sensitive information.



5. Mitigation Measures to address Issues related to Organisation Privacy

5.2 Defend data, devices, and networks from cyber attacks

- Maintain a **clean machine**: the latest security tools, Web browser, and operating system are the greatest defenses against viruses, ransomware, and other online attacks.
- Set your **antivirus** program to **run a scan** after each download.
- Other **critical software updates** should be applied as soon as they are available. However, companies need to have proper software updates testing before deployment. (Latest news: An update from cybersecurity firm CrowdStrike, the source of Microsoft's flaw , 19 July 2024)



5. Mitigation Measures to address Issues related to Organisation Privacy

5.3 Protect Internet access with a firewall

- A firewall is a collection of applications that work together to protect data on a private network from unwanted access.
- Make sure your operating system's **firewall is switched on**.
- If the employees work from home, ensure that their computers are protected by a firewall.



5. Mitigation Measures to address Issues related to Organisation Privacy

5.4 Make a strategy for dealing with handheld devices

- Mobile devices, especially those that hold sensitive data or have access to the business network, may cause significant security and management issues.
- Require users to **password-protect** their smartphones, **encrypt** their data, and install **security software** to prevent hackers from obtaining information when on public networks.
- Set up procedures for **identifying equipment** that has gone missing or has been stolen.



5. Mitigation Measures to address Issues related to Organisation Privacy

5.5 Make copies of critical company records and documents as backups

- Make copies of **critical data** include word-processing records, computer spreadsheets, directories, accounting files, human resources files, and accounts receivable/ payable files.
- Backup **data automatically** or at least on a **regular basis**, as needed, and store copies elsewhere or in the cloud.



5. Mitigation Measures to address Issues related to Organisation Privacy

5.6 Build user profiles for each employee and restrict physical access to the machines.

- Only **trusted IT workers and key personnel** should be granted administrative rights view or use company machines.
- **Laptops** are especially vulnerable to theft or loss, so keep them **locked up** while not in use.
- Make sure that each employee has his or her own user account and that **good passwords** are used.



5. Mitigation Measures to address Issues related to Organisation Privacy

5.7 Keep the Wi-Fi networks secure.

- Make sure your workplace's **Wi-Fi network** is safe and secure.
- To mask your Wi-Fi network, configure your wireless access point or router to **not broadcast the network name**, also known as the Service Set Identifier (SSID).
- The router's access is **password protected**.



5. Mitigation Measures to address Issues related to Organisation Privacy

5.8 Authentication and passwords

- Employees are required to use **strong passwords** and **update** them every three months.
- Consider using **multifactor authentication**, which needs more detail than just a password to obtain access.



5. Mitigation Measures to address Issues related to Organisation Privacy

5.9 Internet Filtering

- To protect web users, network and devices from various online threats (eg malware, phishing websites), Internet filtering is used.
- Internet filtering can be based on the following criteria to be included either in a **blacklist** or **whitelist**:
 - URLs
 - Domain
 - Keywords
 - Content
 - Category



5. Mitigation Measures to address Issues related to Organisation Privacy

5.9 Internet Filtering (Cont.)

- There are many types of Internet filters solutions for companies or consumers to choose from based on requirements.
- Internet Filters can be installed as a **web browser extension**, a **standalone program** or as **part of total security solutions**.
- Another type of Internet Filter is by using **search engine filters**, which restricts SERPs (Search Engine Result Pages) listing to avoid malicious sites. For example, Google SafeSearch.
- Internet Filters can be installed by consumers, companies or Internet Service Provider (ISP)



Question?

"Ethics is knowing the difference between what you have a right to do and what is right to do." - Potter Stewart



Disclaimer:

The slide content has been sourced from the following reference books and Internet resources. Refer to the following list for more details and further reading:

1. Awari, G.K., & Warjekar, S.V. (2022). *Ethics in Information Technology: A Practical Guide*. CRC Press, Taylor & Francis Group.
2. Miranda, Gonçalves, Rubén., et al. Artificial Intelligence and Human Rights, Dykinson, S.L. (2021). ProQuest Ebook Central. <http://ebookcentral.proquest.com/lib/tarc-ebooks/detail.action?docID=6851958>

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